

GodownPro

Material & Project Management System

Yorks Airconditioning and Refrigeration LLC SPC — Abu Dhabi, UAE

Type	Roles	Stack
Final MVP Specification	Engineer · Procurement · Admin	Flutter · Firebase

1. Company & Purpose

Yorks Airconditioning and Refrigeration LLC SPC is a full-service HVAC contractor based in Abu Dhabi, UAE. They cover the complete lifecycle of HVAC projects — from initial site inspection and material planning, through procurement and supply, to full installation and commissioning.

GodownPro is built to eliminate the uncertainty and inefficiency in their current material and project management process. Today, material requests are handwritten, stock visibility is near zero for site engineers, and there is no clean record of who requested what, when it was approved, or what was actually dispatched. This results in delays on site, material going missing with no audit trail, and projects running over budget with no data to explain why.

GodownPro solves this with two apps on one Firebase backend — a Flutter mobile app for engineers on site, and a Flutter web app for the procurement team and admin in the office.

2. The Three Roles

Role	Platform	What they do
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Engineer انجینئر	Mobile App (iPhone / Android)	Creates projects, plans material requirements, submits day-to-day material requests during execution, reviews and approves what procurement has arranged
Procurement پروکیورمنٹ	Web App (Desktop browser)	Receives all requests, comments and discusses with engineer, auto-checks inventory, arranges external items, dispatches materials, maintains inventory stock levels
Admin ایڈمن	Web App (Desktop browser)	Full system control — manages users, inventory, projects, approves or deletes any request, revokes access, views complete audit trail of every action in the system

There are only three roles in the system. No delivery guy app. Procurement is responsible for physical dispatch and delivery — they give the driver the location and items list outside the system.

3. The Two Phases of Every Project

Every project in GodownPro goes through two distinct phases. Phase 1 happens before work starts on site. Phase 2 happens during the execution of the work. Both phases are tracked, logged, and tied to the same project.

Phase 1 — Planning Before work starts on site	Phase 2 — Execution During active site work
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Engineer creates the project with location, building, floor details	Engineer requests items as needed from site (fast flow)
Engineer lists the major items expected to be needed (rough plan, editable)	Can request any item — from inventory or as a custom item
Procurement reviews, auto-checks inventory, sources external items	Procurement sees request and dispatches immediately if in stock
Back-and-forth comments between Engineer and Procurement	If not in stock: procurement decides partial or hold
Final approval from Engineer before project is activated	Stock auto-deducts on every dispatch

4. Phase 1 Workflow — Project Planning (Step by Step)

This phase runs once per project, before any site work begins. The goal is to get the material plan agreed between the Engineer and Procurement so everything is arranged and ready.

#	Who	Action	Detail
1	Engineer	Create Project	Opens app, creates new project. Fills in: Project Name, Client, Location, Building Name, Floor Number(s), Start Date, and any relevant site notes.
2	Engineer	Add Material Plan	Lists all major items expected for the project. For each item selects from inventory (auto-fills fields) or creates a custom item with: Description, Brand, Country of Origin, Size, Quantity, Unit, RAL Colour. This is a rough plan — not a final order.
3	Engineer	Submit to Procurement	Taps Submit. Procurement receives an in-app notification immediately.
4	Procurement	Review & Auto-Check	Opens the plan. Items that exist in inventory are automatically green-ticked. Items not in stock are flagged for external sourcing. Procurement can comment on any item asking for clarification.
5	Both	Comment & Discuss	Back-and-forth comments between Engineer and Procurement on specific items. Engineer can edit the plan at this stage — change quantities, swap items, add items they missed.

6	Procurement	Arrange External Items	Procurement sources and arranges all items not in stock. Marks each item as Arranged when done. When all items are arranged or confirmed from stock, clicks Done.
7	Engineer	Review Arranged Plan	Engineer sees the full list — what was arranged, what stock it came from, any notes from procurement. If satisfied, clicks Approved. If not, rejects specific items with a comment explaining what needs to change.
8	Both	Re-approval if Changes	If engineer rejects any items, procurement revises. If engineer edits the plan AFTER procurement has already marked items done, all changed items are highlighted with a diff view and the approval cycle restarts for those items only.
9	Engineer	Final Approval	Engineer clicks Approved. Project status changes to Active. Phase 2 can now begin.

5. Phase 2 Workflow — Execution Requests (Step by Step)

This phase runs continuously during site work. The engineer can make multiple requests per day. The flow is intentionally fast — no extra approval gates. Procurement owns dispatch end to end.

#	Who	Action	Detail
1	Engineer	Open Active Project	Opens the project from their app. Sees the Phase 1 plan and all previous Phase 2 requests for context.
2	Engineer	Create Material Request	Taps New Request. Selects items from the inventory picker (search + category filter) OR creates a custom item using the standard fields. Sets quantity for each item. Adds an urgency flag and note if needed.
3	Engineer	Submit Request	Taps Submit. Procurement receives an in-app notification immediately. It is procurement's responsibility to act on this without delay.

4	Procurement	Review & Dispatch	Sees the request. If items are in stock — dispatches immediately. Stock auto-deducts. Engineer is notified. If items are partially in stock — procurement decides: dispatch available stock now and source the rest, or hold the full request. Procurement uses their judgement.
5	Procurement	Handle Out-of-Stock Items	For items not in stock, procurement arranges externally and updates the request when the item is ready. Engineer is notified at each status change.
6	Engineer	Receive & Confirm	Engineer sees their request updated with what was dispatched, quantity issued, and time. Full visibility on site of exactly what is coming.

6. Edit & Re-Approval Rules

Situation	What Happens
Engineer edits plan BEFORE procurement has acted	Free edits allowed. No re-approval needed. Just save and continue.
Engineer edits plan WHILE procurement is reviewing	Changes are flagged with a highlight. Procurement sees a diff of what changed. They must re-review the changed items.
Engineer edits plan AFTER procurement has clicked Done	Changed items are highlighted in a diff view. Procurement must re-arrange or re-confirm the changed items. Only the changed items go through re-approval — not the whole plan.
Procurement edits quantities or swaps items	Engineer is notified. The changed items show a diff. Engineer must re-approve those specific items before project goes Active.
Phase 2 request — engineer wants to change after submitting	Can only be cancelled before procurement has acted. Once procurement has dispatched, a new request must be created for any additional items.

7. Custom Item Fields

When an engineer cannot find an item in the inventory during either Phase 1 planning or a Phase 2 request, they can create a custom item. Custom items use the same standard fields as the inventory to ensure consistency and give procurement enough information to source correctly.

Field	Required?	Example	Notes
Material Description	Required	Special GI Bracket for 6 inch pipe	Clear descriptive name
Brand / Supplier	Optional	Hilti, Local	If engineer knows the brand they want
Country of Origin	Optional	UAE, Germany	If project specs require a specific origin
Size	Required	150mm x 80mm	Dimensions, model code, or spec
Quantity	Required	10	How many needed
Unit	Required	Nos, Mtr, Kg, Roll	Unit of measurement
RAL Colour	Optional	RAL 9006, White, Ivory	Required for grilles, dampers, inlet/outlet items
Note to Procurement	Optional	Urgently needed for 3rd floor ceiling	Free text for anything not covered above

Custom items created by engineers are reviewed by procurement. If procurement sources and adds the item to the inventory, it becomes a standard inventory item available to all engineers on future requests. Admin can also add items directly to inventory.

8. Inventory Management

The inventory is the single source of truth for all materials held in the Yorks godown. Procurement owns and maintains it. Stock levels auto-deduct on every Phase 2 dispatch. The inventory is visible (read-only) to engineers so they can see what is available before making requests.

Inventory Item Fields

Col umn	Field	Example	Notes
A	Material Description	HDGAF Cable Tray Epoxy 7032	Full descriptive name

B	Brand / Supplier	DELTA, TMI, Honeywell	Brand or supplier name
C	Country of Origin	UAE, USA, Japan	For compliance and project specs
D	Size	600X50X2X3, 3 1/8", 32mm	Dimensions or spec code
E	Quantity	120	Auto-deducts on dispatch
F	Unit	LTH, Nos, Mtr, Kg, Roll, Drum	Unit of measurement
G	RAL Colour	RAL 9006, White, Ivory, Alum.	Mandatory for inlet/outlet items
H	Unit Cost (AED)	194.00	Visible to Admin and Accountant only
I	Category	Ducting, Copper Pipe, Valves...	28 categories from stock list
J	Store Location	Ground Floor Store 1, Shelf E-2	Physical location in godown

Stock Status Badges

Badge	Colour	Meaning
In Stock	Green	Sufficient quantity available for the requested amount
Low Stock	Amber	Quantity available but below a minimum threshold — engineer is warned
Out of Stock	Red	Zero quantity — item cannot be selected in a fast Phase 2 request, must be custom requested
Auto-ticked	Green tick	Item in a Phase 1 plan that already exists in inventory with sufficient stock

9. Admin — Full System Control

There is one Admin in the system. The Admin has full unrestricted access to every part of the application. No action in the system can override or bypass the Admin.

Admin Capability	Detail
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User Management	Create user accounts, assign roles (Engineer or Procurement), reset passwords, revoke access immediately if needed
Inventory Control	Add, edit, or delete any inventory item. Set unit costs. Adjust stock levels manually if needed (e.g. after physical stocktake)
Project Control	View all projects in all phases. Delete any project. Reopen a closed project if needed
Request Control	View every request in the system. Approve or delete any request at any stage regardless of its current status
Access Permissions	Control which engineers can view which projects. Grant or revoke store inventory view access per user
Full Audit Trail	See a complete timestamped log of every action taken by every user — who created what, who approved what, who dispatched what, and when. Nothing is hidden
Notifications	Receives in-app notification for any critical event — stuck requests, disputes, high-value dispatches

The audit trail is non-deletable. Even the Admin cannot delete a log entry. Every action in the system — create, edit, approve, reject, dispatch, delete — is permanently recorded with the actor's name, role, and exact timestamp.

10. Notification Rules

In-app notifications are used for MVP. WhatsApp integration is planned for a later phase. Notifications are real-time via Firestore streams — no delay.

Trigger	Who Gets Notified	Message
Engineer submits Phase 1 plan	Procurement	New project plan submitted — [Project Name]
Procurement comments on an item	Engineer	Procurement has a question on [Project Name]
Procurement clicks Done	Engineer	Your material plan is ready for review — [Project Name]

Engineer approves Phase 1	Procurement	Phase 1 approved — [Project Name] is now Active
Engineer rejects Phase 1 items	Procurement	Engineer has requested changes on [Project Name]
Engineer edits plan after Procurement Done	Procurement	Plan updated — [X] items changed in [Project Name]
Engineer submits Phase 2 request	Procurement	New material request from [Engineer Name] — [Project Name]
Procurement dispatches items	Engineer	Your request has been dispatched — [Project Name]
Procurement partially dispatches	Engineer	Partial dispatch — [X] of [Y] items sent — [Project Name]
Any request stuck for 24 hours	Admin	Unactioned request alert — [Project Name]

11. Project Creation Fields

When an engineer creates a new project in Phase 1, the following details are captured. These fields appear on every request and report tied to the project.

Field	Required?	Example
Project Name	Required	Al Barsha Tower B — HVAC Installation
Client Name	Required	Aldar Properties
Location / Area	Required	Al Barsha, Dubai
Building Name	Required	Tower B, Block 3
Floor Number(s)	Required	Floors 3, 4, 5
Site Address	Optional	Sheikh Zayed Road, Al Barsha
Start Date	Required	15 Jan 2026
Expected End Date	Optional	30 Mar 2026
Site Notes	Optional	Access via Gate 2 only. Contact Ahmed on site.

12. Screens by Role

Flutter Mobile App — Engineer

- Login — email and password, no signup, credentials set by Admin
- Home — list of my active projects with phase status badge
- Create Project — form with all project fields
- Phase 1 Plan — item list, add from inventory or custom, submit to procurement
- Item Picker — search bar, category filter chips, inventory list with stock badges, quantity stepper
- Custom Item Form — all standard fields for items not in inventory
- Plan Review — procurement's arranged list, comment thread, approve or reject
- Diff View — highlights changed items when re-approval is triggered
- Phase 2 Request — fast item picker, optional urgency flag, submit
- Request Status — what was dispatched vs requested, timestamps
- Notifications — in-app notification centre

Flutter Web App — Procurement

- Login — email and password
- Dashboard — pending Phase 1 plans, pending Phase 2 requests, low stock alerts
- Phase 1 Queue — list of plans awaiting review
- Plan Detail — item list, auto-ticked inventory items, comment thread, mark items arranged, click Done
- Phase 2 Queue — incoming live requests, sorted by urgency and time
- Dispatch Screen — confirm quantities, dispatch, partial dispatch option
- Inventory Management — add, edit, update stock levels, view all 28 categories
- Add Stock Item — full form with all inventory fields
- Issuance History — log of all dispatched items

Flutter Web App — Admin

- Dashboard — full system overview, all active projects, pending requests, total stock value
- User Management — create users, assign roles, revoke access
- All Projects — every project, every phase, with ability to delete
- All Requests — full searchable log of every request across all projects
- Inventory — full control, add/edit/delete items, adjust stock manually
- Audit Trail — complete timestamped log of every action by every user
- Notifications — critical alerts for stuck requests and escalations

13. Firebase Data Model (Firestore)

Eight Firestore collections cover the complete system. Real-time listeners on materialRequests and notifications ensure all users see updates instantly without refreshing.

collection: `users`

Field	Type	Notes
<code>uid</code>	string	Firebase Auth UID — document ID
<code>name</code>	string	Full display name
<code>role</code>	enum	engineer procurement admin
<code>assignedProjects</code>	array	Project IDs visible to this engineer
<code>canViewInventory</code>	boolean	Admin-controlled read access to inventory
<code>createdAt</code>	timestamp	Account creation time
<code>createdBy</code>	string	Admin UID who created this account

collection: `projects`

Field	Type	Notes
<code>id</code>	string	Auto-generated
<code>name</code>	string	Project name
<code>clientName</code>	string	Client company
<code>location</code>	string	Area / address
<code>buildingName</code>	string	Building name
<code>floors</code>	string	Floor numbers covered
<code>startDate</code>	timestamp	
<code>endDate</code>	timestamp	Optional
<code>siteNotes</code>	string	Optional notes
<code>phase</code>	enum	planning active completed on_hold
<code>createdBy</code>	string	Engineer UID
<code>createdAt</code>	timestamp	
<code>updatedAt</code>	timestamp	

collection: `materialPlans` (Phase 1)

Field	Type	Notes
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id	string	Auto-generated, one per project
projectId	string	Reference to projects
items	array	[{ id, description, brand, origin, size, qty, unit, ralColour, isCustom, status: pending arranged ticked }]
status	enum	draft submitted procurement_review pending_engineer_approval approved rejected
comments	array	[{ authorId, authorName, authorRole, text, itemRef?, timestamp }]
submittedAt	timestamp	When engineer first submitted
approvedAt	timestamp	When engineer gave final approval
version	number	Increments on every edit — used to show diff
lastEditedBy	string	UID of last person to edit

collection: `materialRequests` (Phase 2)

Field	Type	Notes
id	string	Auto-generated
projectId	string	Reference to projects
requestedBy	string	Engineer UID
items	array	[{ inventoryId?, description, brand, size, qty, unit, ralColour, isCustom, qtyDispatched, status }]
status	enum	pending dispatched partial sourcing cancelled
urgency	enum	normal urgent critical
engineerNote	string	Optional note from engineer
procurementNote	string	Note added by procurement on dispatch
createdAt	timestamp	
dispatchedAt	timestamp	When first dispatch was made

collection: `inventory`

Field	Type	Notes
id	string	Auto-generated

<code>description</code>	string	Material description (Column A)
<code>brand</code>	string	Brand / Supplier (Column B)
<code>countryOfOrigin</code>	string	Country of Origin (Column C)
<code>size</code>	string	Size / spec (Column D)
<code>qtyInStock</code>	number	Current stock — auto-deducts on dispatch (Column E)
<code>unit</code>	string	LTH, Nos, Mtr, Kg, Roll, Drum, Sheet, Cly (Column F)
<code>ralColour</code>	string	RAL Colour — blank if not applicable (Column G)
<code>unitCostAED</code>	number	Unit price AED — Admin and Accountant only (Column H)
<code>category</code>	string	One of 28 categories (Column I)
<code>storeLocation</code>	string	Physical location in godown (Column J)
<code>minStockLevel</code>	number	Threshold for low stock alert
<code>updatedAt</code>	timestamp	Last updated
<code>updatedBy</code>	string	UID of who last updated this item

collection: `dispatches`

Field	Type	Notes
<code>id</code>	string	Auto-generated
<code>requestId</code>	string	Phase 2 request reference
<code>projectId</code>	string	Project reference
<code>dispatchedBy</code>	string	Procurement UID
<code>items</code>	array	[{ inventoryId, description, qtyDispatched, unit }]
<code>dispatchedAt</code>	timestamp	Exact dispatch time
<code>note</code>	string	Optional procurement note

collection: `notifications`

Field	Type	Notes
<code>id</code>	string	Auto-generated
<code>toUserId</code>	string	Recipient UID

<code>type</code>	string	plan_submitted procurement_done engineer_approved request_dispatched etc.
<code>title</code>	string	Short notification title
<code>body</code>	string	Full notification message
<code>refId</code>	string	ID of related project or request
<code>read</code>	boolean	false until user opens it
<code>createdAt</code>	timestamp	

collection: `activityLog`

Field	Type	Notes
<code>id</code>	string	Auto-generated
<code>type</code>	string	project_created plan_submitted item_arranged phase1_approved request_dispatched stock_updated user_created access_revoked etc.
<code>actorId</code>	string	UID of user who performed the action
<code>actorName</code>	string	Denormalised name for fast display
<code>actorRole</code>	string	engineer procurement admin
<code>projectId</code>	string	Related project if applicable
<code>refId</code>	string	Related document ID
<code>detail</code>	string	Human-readable English description of what happened
<code>timestamp</code>	timestamp	Exact time — cannot be deleted or modified by anyone

14. Tech Stack

Technology	Why chosen
Flutter (Web + Mobile)	Single Dart codebase delivers both the mobile app for engineers and the web app for procurement and admin. One developer can maintain both.
Firebase Authentication	Email/password login only. No self-signup anywhere. Admin creates all accounts manually. Role is stored in Firestore on the user's UID document.

Cloud Firestore	Real-time NoSQL database. Status changes — plan approved, request dispatched — appear on every connected device instantly via Firestore streams. No polling.
Firebase Security Rules	Role-based access enforced server-side. Engineers cannot write to inventory. Procurement cannot access the admin panel. These rules cannot be bypassed from the app.
Firebase Hosting	Single command deploy for the web app. CDN-backed global delivery. Custom domain support for Yorks (e.g. app.yorksac.ae).
flutter_riverpod	State management. Reactive providers keep the UI in sync with live Firestore streams across all screens simultaneously.
go_router	Declarative routing with role-based guards. After login the app reads the user's role and routes them to the correct home screen automatically.
easy_localization	English and Urdu/Hindi language switching. Translation JSON files applied across all screens from day one.

15. MVP Scope

✓ Included in MVP	✗ Post-MVP
✓ 3 roles: Engineer, Procurement, Admin	✗ WhatsApp notifications
✓ Phase 1 planning with comment thread	✗ PDF material plan / dispatch slip
✓ Auto inventory green-tick in Phase 1	✗ Barcode / QR scanning for stock
✓ Diff view on re-approval	✗ Low stock automatic alerts
✓ Phase 2 fast request flow	✗ External purchase order module
✓ Custom item creation with standard fields	✗ Supplier / vendor management
✓ Inventory with all 10 columns	✗ Multi-warehouse support
✓ Stock auto-deduct on dispatch	✗ Offline mode for poor signal sites
✓ In-app real-time notifications	✗ Advanced analytics and charts
✓ Admin full control panel	✗ ERP / accounting integration
✓ Non-deletable audit trail	✗ Client-facing portal
✓ English + Urdu/Hindi UI	✗ Automated reporting

16. Roadmap

Phase	Name	Deliverables
MVP	Core system	Full 2-phase workflow, 3 roles, inventory, real-time notifications, audit trail, bilingual UI
v1.1	Notifications + slips	WhatsApp notifications at every step. Printable PDF dispatch slip replacing the handwritten paper.
v1.2	Purchase orders	When items are out of stock, procurement raises a PO from the app. Supplier list. PO tracking and approval.
v1.3	Stock intelligence	Low stock automatic alerts. Minimum stock thresholds per item. Reorder suggestions for procurement.
v2.0	Scale up	Offline mode for poor-signal sites. Barcode/QR scanning for fast stock updates. Advanced analytics for admin. Multi-site inventory.